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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION NOVEMBER 2024
First Semester

Discipline Specific Course - **MG1DSCZIM102-FUNDAMENTALS OF BIOCHEMISTRY**
(2024 Admission onwards)

Duration: 1.5 hrs Max Marks: 50

Part A
Fill in the blanks. Answer **All** question
Each question carries **1** marks

1.Theory of special creation states that life was created by ……. [U] [1]
Ans: **God**
2.……..was the first scientist to view living cell under microscope [U] [1]
Ans: **Antonie van leeuwenhoek**
3. pH of the blood is ….……..
Ans:**7.35-7.45** [A] [2]
4.Buffer resist change in …….when small amount of acid or base are added [K] [2]
Ans :**pH**
5.……..membrane proteins embedded with in the phospholipid bilayer
Ans :**Integral** [U] [3]
6.…… transport is the movement of molecules across a cell membrane without using energy [U] [3]
Ans :**Passive Transport**
7.Uniport is the movement of one rtype of molecules in …….direction
Ans: **One** [U] [4]
8.Anetwork of membraneous tubules and cisternae within the cytoplasm of eukaryotic cell is called……
Ans: **Endoplasmic Reticulum** [U] [4]
9.Photolysis of water occurs in…….photophosphorylation
Ans: **Non cyclic** [K] [5]

10.The first stable product of C4 cycle is……
Ans : **Oxalo Acetic acid** [K] [5]

(10 x 1 =10 Marks)

Part B
Short Questions. Answer any **10** questions
Each question carries **2 Marks**

11.Write short note on orgin of life? [A] [1]
Orgin of life is also called protobiogenesis.Origin of life was explained by theories such as:
1. Theory of special creation

2. Theory of spontaneous generation
3. Theory of biogenesis

12 .State Theory of special creation [K] [1]

According to this theory all the living beings on the earth was created by God or by some supernatural power.

13.State theory of biogenesis [K] [1]

The theory of biogenesis states that living things can only reproduce from other living things, and that life does not arise from non-living material.

14.Explain Louis Pasteur Swan neck Flask Experiment [U] [1]

Pasteur's experiment consisted of boiled broth in flasks with S-shaped necks. The broth was boiled after being poured into the flasks to destroy any living microorganisms that may have been present in the broth. The S-shaped necks prevented microorganisms from entering and contaminating the broth after boiling.

15.Explain the difference between strong and weak acids. [U] [2]

A strong acid is an acid which is completely ionized in an aqueous solution. A weak acid is an acid that ionizes only slightly in an aqueous solution.

16. Describe pH Scale [U] [2]

A pH scale is a tool for measuring acids and bases. The scale ranges from 0-14.pH 7 indicate neutral solution. pH less than 7 indicate acidity, whereas a pH of greater than 7 indicates a base.

17.Describe amphitropic proteins [U] [3]

Amphitropic proteins are a class of proteins that reversibly bind to the hydrophobic part of membranes and lipids.

18.Write short note on simple diffusion [K] [3]

Simple diffusion is the process by which molecules move from an area of high concentration to an area of low concentration without the help of transport proteins.

19.Write short note on simple diffusion [K] [4]

Simple diffusion is the process by which molecules move from an area of high concentration to an area of low concentration without the help of transport proteins.

20..Define Osmosis [K] [4]

Osmosis is the process by which water molecules move through a semipermeable membrane from a region of low solute concentration to a region of high solute concentration.

21 .Write short note on cyclic photophosphorylation [A] [5]

Cyclic Photophosphorylation is a light-dependent reaction in which the high energy electrons are released in a cyclic pathway from P700 to PS-I.

22.Explain the role of RuBisCo in Carbon fixation. [U] [5]

It catalyses the first step (carboxylation)of carbon fixation in the C3 pathway .

(10 x 2 = 20 Marks]

Part C

Short Essay Type Questions. Answer any **5 Q**uestions
Each question carries **4** marks

23. Define biogenesis and explain its significance in understanding life's origin
(Biogenesis definition -1 Mark,Swan neck flask experiment -1 Marks,Significanc-2 Marks)

Biogenesis refers to the process wherein life arises from similar life forms. Significance

1.Life spontaneously arising from nonliving matter

2.Life arising on Earth through a series of chemical reactions [K] [1]

24.Discuss the scope of biochemistry in modern biology (Explain any 4 scopes-1 mark each)

Biochemistry has a broad scope and many applications

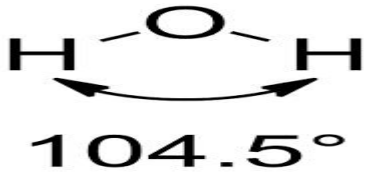
Medicine:Work in the field of drug development, immunology, pathology, pharmacy, and vaccine development.

Agriculture: Biochemists study the biochemistry of crops and medicinal plants. They can work as agricultural scientists.

Biotechnology: Biochemists study how molecules like proteins, nucleic acids, lipids, vitamins, and hormones function.

Nutrition: Biochemists study how essential nutrients like minerals, proteins, carbohydrates, fats, and vitamins affect the human body. [U] [1]

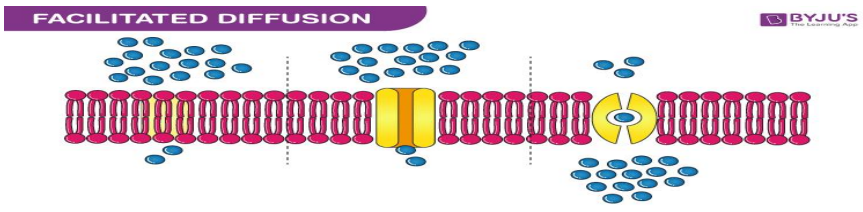
25.Draw structure of water molecule with their bond angle



[K] [2]

26 .Explain the mechanism of facilitated diffusion with diagram(Facilitated diffussion definition-2 Marks,Mechanism-2 Marks)

Facilitated diffusion is the passive movement of molecules along the concentration gradient. The mechanism of facilitated diffusion is as follows:



[U][3]

27.Describe the process of Osmosis
(Osmosis definition -2 Marks , Types -2 Marks)

28.Osmosis is a process by which the molecules of a solvent pass from a solution of low concentration to a solution of high concentration through a semi-permeable membrane.An **isotonic solution** is one that has the same concentration of solutes both inside and outside the cell.A **hypertonic solution** is one that has a higher solute

concentration outside the cell than inside.A **hypotonic solution** is one that has a higher solute concentration inside the cell than outside.**Endosmosis**– When a substance is placed in a hypotonic solution, the solvent molecules move inside the cell and the cell becomes turgid or undergoes deplasmolysis. **Exosmosis**– When a substance is placed in a hypertonic solution, the solvent molecules move outside the cell and the cell becomes flaccid or undergoes plasmolysis. [U] [4]

28.Contrast between Symbiotic and Non symbiotic Nitrogen fixation (Symbiotic Nitrogen fixation-2 Marks,Non symbiotic-2 Marks)

	Symbiotic	**Non-symbiotic**
Relationship with plants	Intimate association with host plants.	Free-living, or non-symbiotic, in the soil
Examples	Rhizobium, Frankia	Cyanobacteria, Azotobacter,
Process	Bacteria invade root hairs, multiply, and stimulate root nodule formation	Bacteria bind nitrogen in the soil and convert it to ammonia

[An] [5]

29. Write an essay on larger subunit of nitrogenase enzyme
(Nitrogenase enzyme-1 Marks,Components-2 Marks,Diagram-1 Mark)

Nitrogenase is the key enzyme for biological nitrogen fixation. Nitrogenase converts nitrogen to ammonia.Nitrogenase catalyses the conversion of nitrogen to ammonia. Nitrogenases are made up of two components.

Component I – The first component or component I is known as dinitrogenase. It is Mo-Fe protein (or V-Fe or Fe-only). It contains P clusters and two Fe-Mo cofactors within alpha subunits. [A] [5]

(5 X 4= 20 Marks)

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